COMPTIA SECURITY+ SY0-701 | DOMAIN 3 · SECURITY ARCHITECTURE (18%)

Security+ 3.1 · Network Infrastructure Concepts

Mestre CyberSec · **Lesson 3.1 — Network Infrastructure Concepts** (6:56) · Assistir Vídeo no YouTube ↗ · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: Segmentação: VLAN (lógica no mesmo hardware); SDN (software-defined networking — control plane separado do data plane; flexível e programável); microsegmentação (isola carga por carga — Zero Trust). VPC (rede privada virtual na nuvem, isolada).

💡 Key Exam Highlights & Critical Concepts

- SDN: control plane centralizado; regra programável e dinâmica
- Microsegmentação: isola cada workload — Zero Trust aplicado em rede
- VPC: tua rede isolada do resto do provider dentro da nuvem

🎯 Exam Tips & Typical CompTIA Scenarios

- 'centralized control plane managing network routing' = SDN
- 'isolates individual workloads' = microsegmentação

🚨 Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 3.1 — Network Infrastructure Concepts)

Why is this advanced coverage required for SY0-701?: Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Virtual Local Area Network (VLAN)

A logical segmentation of a physical network that groups devices into separate broadcast domains on the same hardware, improving security and reducing broadcast traffic.

⚠️ ***Exam Trap:*** *If the question mentions 'logical segmentation on the same physical switch' or 'separating broadcast domains,' the answer is VLAN — not subnet.*

• Software-Defined Networking (SDN)

A network architecture that separates the control plane (routing decisions) from the data plane (traffic forwarding), enabling centralized, programmable, and dynamic network management.

⚠️ ***Exam Trap:*** *Phrases like 'centralized control plane managing routing' or 'programmable network rules' point directly to SDN — do not confuse with traditional VLAN routing.*

• Control Plane vs Data Plane (N/A)

The control plane makes routing and policy decisions, while the data plane (forwarding plane) actually moves the packets. SDN decouples these for flexibility.

⚠️ ***Exam Trap:*** *Any exam question mentioning 'separating routing decisions from traffic forwarding' describes the SDN control/data-plane model.*

• Microsegmentation (N/A)

A security approach that isolates each individual workload, application, or service with its own security policy, applying Zero Trust principles at the network level.

⚠️ ***Exam Trap:*** *Trigger phrase 'isolates individual workloads' = microsegmentation. Distinct from VLAN, which segments by group/subnet, not per-workload.*

• Virtual Private Cloud (VPC)

An isolated virtual network within a public cloud provider, logically separated from other tenants and other networks on the same cloud provider.

⚠️ ***Exam Trap:*** *If the question describes 'your isolated network inside a cloud provider,' think VPC — it is the cloud equivalent of a private network, not a traditional VPN tunnel.*

• Zero Trust (Network Context) (ZT)

A security model that assumes no implicit trust, requiring verification for every device, user, and workload — frequently enforced through microsegmentation.

⚠️ ***Exam Trap:*** *When a question mentions 'never trust, always verify' combined with 'workload-level isolation,' the answer combines Zero Trust with microsegmentation.*

• Network Segmentation (N/A)

The practice of dividing a network into smaller zones (using VLANs, subnets, or SDN) to limit lateral movement and contain security breaches.

⚠️ ***Exam Trap:*** *The umbrella term. If the question describes 'containing breaches' or 'limiting lateral movement,' segmentation (broad) is the concept — pick VLAN vs microsegmentation based on granularity.*

• East-West vs North-South Traffic (N/A)

North-South traffic enters/leaves the data center (user to server); East-West traffic flows server-to-server within the data center — microsegmentation primarily secures East-West flows.

⚠️ ***Exam Trap:*** *Questions about 'server-to-server internal traffic' or 'lateral movement' refer to East-West traffic, which is the target of microsegmentation policies.*

🏛️ Conceptual Deep Dive, Real-World Analogies & GRC Analysis

📌 Virtualization and Hypervisors: Type 1 (Bare-metal) vs. Type 2 (Hosted)

🏠 **Real-World Analogy:** Think of foundation construction: A Type 1 Bare-metal hypervisor pours concrete directly onto bedrock (operating directly on bare server silicon with no intermediary host OS). A Type 2 Hosted hypervisor is pitching an indoor tent inside your living room on top of carpet (it runs as a desktop application inside Windows or macOS, like VirtualBox or VMware Workstation).

🔍 **Root Cause & Background:** Type 1 eliminates OS overhead and reduces attack surfaces, serving as the enterprise standard for cloud providers (ESXi, Hyper-V, KVM). Type 2 suffers high context-switching latency and inherits all OS vulnerabilities and crash risks of the underlying host desktop.

🛡️ **GRC Auditor Perspective:** Verify that enterprise production virtualization hosts exclusively run Type 1 hypervisors; validate VM escape hardening benchmarks and ensure hypervisor management consoles are isolated from production networks.

⚠️ **How CompTIA Tests This:** Trigger: 'hypervisor runs directly on the bare-metal physical hardware without an underlying host operating system'. Answer: Type 1 Hypervisor (Bare-metal). If 'runs as software application inside host OS' -> Type 2 Hypervisor.

📌 VPC (Virtual Private Cloud) and Public vs. Private Subnets

🏠 **Real-World Analogy:** A VPC is an entire locked private floor leased exclusively by your enterprise inside a public skyscraper. A Public Subnet is the reception lobby facing the public street entrance where receptionists greet visitors (web servers with public IPs and an Internet Gateway route). A Private Subnet is the inner vault down a secured hallway with no exterior windows or street entrance (database cluster accessible only internally via private IP).

🔍 **Root Cause & Background:** Network boundary isolation ensuring sensitive persistence tiers (databases, backend APIs) are physically unreachable from the public internet, preventing automated port scanning and unauthenticated brute-force probing.

🛡️ **GRC Auditor Perspective:** Audit VPC route tables to ensure private subnets have zero inbound routes to Internet Gateways; enforce NAT Gateway egress architecture for isolated package updates and mandate NACLs/Security Groups blocking public CIDRs.

⚠️ **How CompTIA Tests This:** Trigger: 'secure database tier must receive security updates from internet but cannot accept any inbound internet connections'. Answer: Private Subnet with NAT Gateway.

📌 Multi-Network Interconnection: Transit Gateway vs. VPC Peering

🏠 **Real-World Analogy:** Imagine interconnecting 20 corporate offices: VPC Peering is stringing dedicated point-to-point cables between every single building pair (a chaotic spiderweb requiring 190 separate links). Transit Gateway is installing a centralized grand junction roundabout: every office connects once to the hub, which routes traffic dynamically to any authorized destination.

🔍 **Root Cause & Background:** VPC peering lacks transitivity (peering A-to-B and B-to-C does not route A-to-C). Scale limits and routing spaghetti force organizations to deploy Transit Gateways as centralized hub-and-spoke routers enabling consolidated firewall inspection.

🛡️ **GRC Auditor Perspective:** Validate hub-and-spoke transit gateway topologies; confirm all cross-VPC traffic traverses an inspection VPC equipped with centralized NGFW firewalls and aggregated VPC Flow Logs.

⚠️ **How CompTIA Tests This:** Trigger: 'centralized hub-and-spoke cloud router connecting dozens of VPCs and on-premises data centers without complex mesh peering'. Answer: Transit Gateway.

📌 Secure Hybrid Connectivity: IPsec VPN vs. Direct Connect / ExpressRoute

🏠 **Real-World Analogy:** An IPsec VPN is driving an armored bank truck over public highway lanes: your cargo is encrypted and shielded, but rush hour congestion and potholes cause unpredictable transit delays (public internet jitter). Direct Connect or ExpressRoute is laying a private dedicated underground railway connecting your warehouse directly to the depot: completely leased physical fiber with predictable throughput and zero public congestion.

🔍 **Root Cause & Background:** High-throughput enterprise workloads require deterministic latency, SLA guarantees, and massive bandwidth that public internet tunnels cannot deliver. Note that Direct Connect is a private circuit, but requires explicit MACsec or IPsec overlay if end-to-end payload encryption is mandated.

🛡️ **GRC Auditor Perspective:** Audit dedicated private circuits for required encryption-in-transit (MACsec at Layer 2 or IPsec over Direct Connect at Layer 3); test automated failover routing to secondary IPsec VPN tunnels during physical line cuts.

⚠️ **How CompTIA Tests This:** Trigger: 'dedicated, private physical network connection between on-premises datacenter and cloud provider offering predictable throughput and low latency'. Answer: Direct Connect / ExpressRoute.

⚡ QUICK-REFERENCE EXAM CHEAT SHEET

Typical Exam Scenario / Practical Situation	Technical Concept & Target Answer
Hypervisor installed directly on bare-metal physical servers without an intermediary host operating system	**Type 1 Hypervisor (Bare-metal)**
Logically isolated software-defined virtual cloud network enclosing instances, subnets, and custom routing tables	**VPC (Virtual Private Cloud)**
Subnet without direct public Internet Gateway routing designed to isolate backend databases and confidential servers	**Private Subnet**
Centralized hub-and-spoke cloud routing service interconnecting dozens of VPCs and corporate on-premises networks	**Transit Gateway**
Encrypted secure tunnel traversing public internet infrastructure to interconnect branch offices to headquarters	**Site-to-Site IPsec VPN**
Dedicated private leased line bypassing the public internet to deliver predictable latency and high throughput to cloud	**Direct Connect / ExpressRoute**

✅ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A company uses one Layer 2 switch to connect Finance workstations and guest Wi-Fi access points. The security team wants Finance traffic to be logically separated from guest traffic and to restrict guest access to Finance resources without purchasing additional switches. Which solution best meets the requirement?

A) Configure VLANs for Finance and Guest, then apply ACLs on a Layer 3 device to control inter-VLAN traffic.

B) Create a separate VPC for each department inside the on-premises switch.

C) Deploy an SDN controller to replace the need for any logical network separation.

D) Install microsegmentation agents on every guest device to enforce Zero Trust.

QUESTION 2 · EXAM SCENARIO

An enterprise wants routing and policy decisions to be made by centralized software and then pushed programmatically to switches and routers, while the switches continue to forward packets at high speed. Which architecture should the network team implement?

A) VLAN trunking with 802.1Q tagging on all uplinks

B) Software-defined networking with a centralized controller

C) VPC peering between on-premises and cloud networks

D) Network access control using 802.1X posture checks

QUESTION 3 · EXAM SCENARIO

A hospital has its EHR web servers and database servers on the same VLAN. The security team wants to ensure the web tier can reach the database only on the required port and that a compromised web server cannot move laterally to other workloads. Which approach best aligns with Zero Trust?

A) Move all servers into one larger flat VLAN to simplify firewall rules.

B) Place a web application firewall in front of the database servers.

C) Apply microsegmentation policies that allow only required workload-to-workload flows.

D) Create a VPC for each individual server and expose all subnets to the internet.

QUESTION 4 · EXAM SCENARIO

A company is deploying an application in a public cloud. The architect needs an isolated virtual network with its own IP address ranges, subnets, route tables, and security controls. Which cloud construct should be created?

A) VLAN

B) SDN overlay tunnel

C) Microsegmentation zone

D) VPC

QUESTION 5 · EXAM SCENARIO

After a malware infection on an internal virtual machine, the infected VM begins scanning other VMs on the same subnet. The security team wants to stop east-west lateral movement without redesigning the physical network. Which control is most effective?

A) Deploy a web application firewall to inspect inbound HTTP requests.

B) Implement host-based microsegmentation with least-privilege allow lists between workloads.

C) Create a new VLAN for every VM and route all traffic through the internet.

D) Use VPC peering between internal subnets to isolate the infected VM.

QUESTION 6 · EXAM SCENARIO

A SOC is classifying traffic for monitoring. It observes connections from external users on the internet to the organization's public web servers, and responses from those web servers back to the external users. How should this traffic be classified for network visibility purposes?

A) North-south traffic

B) East-west traffic

C) Control plane traffic

D) Overlay traffic

QUESTION 7 · EXAM SCENARIO

A penetration test shows that an attacker connected to an access port can send double-tagged frames and reach a different VLAN. Which configuration change best reduces this VLAN hopping risk?

A) Enable Dynamic Trunking Protocol on all access ports.

B) Place all VLANs into the same native VLAN for simpler trunking.

C) Disable DTP and change the native VLAN on trunks to an unused VLAN ID.

D) Replace the Layer 2 switches with a cloud VPC.

QUESTION 8 · EXAM SCENARIO

A network administrator is documenting how the network separates decisions about the best path from the actual movement of packets. Which statement correctly describes the control plane and data plane?

A) The data plane decides routes, and the control plane forwards packets.

B) The management plane forwards packets, and the data plane authenticates users.

C) The control plane encrypts traffic, and the data plane decides routes.

D) The control plane decides paths, and the data plane forwards packets.

QUESTION 9 · EXAM SCENARIO

A cloud team has a production VPC and a shared-services VPC in the same cloud region. The team needs private communication between them without sending traffic over the public internet. Which VPC feature should be used?

A) VPC peering

B) VLAN trunking

C) SDN controller clustering

D) Microsegmentation agent

QUESTION 10 · EXAM SCENARIO

A company is adopting Zero Trust. A contractor currently connects through a VPN and receives access to the entire internal network, but only needs to use one HR application. Which change best follows Zero Trust network principles?

A) Assign the contractor to the HR VLAN so the contractor can reach all HR resources.

B) Enforce identity-aware microsegmentation that grants access only to the specific HR application.

C) Place the HR application in a public subnet so the VPN is no longer required.

D) Disable the VPN and rely only on the SDN control plane for access decisions.

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option A — Configure VLANs for Finance and Guest, then apply ACLs on a Layer 3 device to control inter-VLAN traffic.

VLANs create logical Layer 2 segmentation on the same physical switch. Inter-VLAN routing can then be controlled with ACLs on a Layer 3 switch or router. VPCs are cloud constructs, SDN does not by itself replace segmentation, and endpoint agents on guest devices are not appropriate for separating broadcast domains and controlling access at the network layer.

Question 2: Option B — Software-defined networking with a centralized controller

SDN separates the control plane, where decisions are made, from the data plane, which forwards traffic. A centralized controller can program network devices through APIs. VLAN trunking is Layer 2 tagging, VPC peering is cloud connectivity, and 802.1X NAC controls admission, not programmable routing.

Question 3: Option C — Apply microsegmentation policies that allow only required workload-to-workload flows.

Microsegmentation applies granular, identity- or workload-based policies inside a segment, limiting lateral movement and supporting Zero Trust. A larger flat VLAN weakens segmentation, a WAF protects web applications from HTTP attacks but not internal database flows, and exposing subnets to the internet is insecure.

Question 4: Option D — VPC

A Virtual Private Cloud is a logically isolated virtual network in a cloud provider, with its own IP ranges, subnets, route tables, gateways, and security controls. VLANs are typically Layer 2 on-premises constructs, an SDN overlay is an architecture, and microsegmentation is a policy technique, not the isolated cloud network boundary.

Question 5: Option B — Implement host-based microsegmentation with least-privilege allow lists between workloads.

East-west traffic is communication between internal workloads. Host-based microsegmentation can enforce per-workload allow lists even when VMs are on the same subnet, reducing lateral movement. A WAF targets inbound web traffic, routing through the internet is inappropriate, and VPC peering connects networks rather than isolating workloads.

Question 6: Option A — North-south traffic

North-south traffic enters or leaves the organizational network boundary, such as internet clients communicating with public web servers. East-west traffic is between internal systems. Control plane traffic carries routing or policy decisions, and overlay traffic is encapsulated traffic in virtual networks.

Question 7: Option C — Disable DTP and change the native VLAN on trunks to an unused VLAN ID.

VLAN hopping can be mitigated by disabling DTP on access ports, explicitly configuring access or trunk modes, and setting trunk native VLANs to an unused VLAN. Enabling DTP or sharing a native VLAN increases risk. A VPC does not replace Layer 2 switch hardening.

Question 8: Option D — The control plane decides paths, and the data plane forwards packets.

The control plane makes routing, switching, and policy decisions. The data plane, also called the forwarding plane, carries user traffic according to those decisions. The other statements reverse or misassign these functions.

Question 9: Option A — VPC peering

VPC peering creates a private network connection between two VPCs so resources can communicate using private IP addresses without internet transit. VLAN trunking is Layer 2 tagging, SDN controller clustering does not connect VPCs, and microsegmentation enforces workload policies rather than VPC-to-VPC connectivity.

Question 10: Option B — Enforce identity-aware microsegmentation that grants access only to the specific HR application.

Zero Trust limits access based on identity and context, not broad network location. Identity-aware microsegmentation can permit only the required application flow for the contractor. A broad HR VLAN still grants excessive access, placing the app in a public subnet removes protection, and SDN alone does not enforce user-to-application least privilege.